Fiscal Year:	FY 2004	**Task Last Updated:**	FY 11/16/2006
PI Name:	Stone, Lee S. Ph.D.		
Project Title:	Oculometric Technologies For Aerospace Applications		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	10/01/2002	**End Date:**	09/30/2006
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Beutter, Brent (NASA Ames Research Center) Adelstein, Bernard (NASA Ames Research Center) McCann, Robert (NASA Ames Research Center) Eckstein, Miguel (University of California, Santa Barbara) Bush, Geoffrey (Lockheed Martin)		
Grant/Contract No.:			
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Performance Goal Text:			

Task Description:	Eye-movement measurements provide a potentially powerful tool for monitoring human performance in tasks involving perceptual and cognitive processing, for controlling computers and other interactive displays, and for understanding the perceptual adaptation to spaceflight. However, there are both technical (i.e. the need for non-intrusive yet precise eye-tracking capabilities) and scientific (i.e., the need to make reliable quantitative inferences about the human operator's perceptual and cognitive state from eye-movement data) requirements for the safe and effective use of eye-movement monitoring in aerospace applications. The proposed study is to perform human-performance research as well as technical feasibility/validation tests to facilitate the incorporation of eye-movement technologies into aerospace applications in support of the Space Human Factors Engineering program. The main goal for the next 4 years will be to validate oculometrics (a technology developed under the previous funding period that allows the quantitative measurement of human perceptual-cognitive performance from eye-movement data alone) over a range of aerospace-related visual tasks, and to extend this technology to other sensory modalities. In addition, we will explore applications of this technology to the development of robust hands-free human-machine interfaces and feedback-systems for training. More specifically, our aims are three-fold. First, we will extend our high-precision non-invasive eye-tracking hardware and software and our oculometric analysis algorithms to head-free and binocular visual conditions and to auditory, vestibular, and stereomotion display conditions. Second, we will use oculometrics to determine quantitatively the correlation between perceptual and/or cognitive state and eye movements during localization, search, and motion-tracking. Third, we will explore initial specific applications of eye-movement monitoring to two aerospace interface prototypes: 1) a speech interface to a robotic assistant to provide robust speech-recognition in high-noise environments and to allow hands-free mode selection, and 2) a low-fidelity, shuttle-landing interface to examine the use of oculomotor feedback for training and performance monitoring. A better understanding of human perceptual and cognitive performance during aerospace tasks has been hampered by the difficulty in making meaningful and precise performance measures without interfering with the task at hand. The need for better performance monitoring and training is even more important when one considers the potential adverse affects of sensorimotor adaptation to microgravity (e.g. emergency egress or landing the shuttle upon return to 1-g). The high-precision eye-tracking technologies validated in our laboratory, together with quantitative oculometrics of vestibular, auditory, and visual perception and of cognitive state, could be used in future flight experiments to help resolve critical human-factors safety issues, and provide concrete guidance in the development and validation of countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report received during this reporting period. [NOTE: Project added toTask Book September 2006 when received information.]
Bibliography Type:	Description: (Last Updated: 10/30/2019)